

# Networking & Communication T02 — Network Foundations

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## 1. What Is a Computer Network?

A **network** is simply a group of devices (computers, phones, printers, servers) connected together so they can **share data and resources**.

### Simple analogy

Think of a **school**:

- Students = devices (phones, laptops)
- Classrooms = networks / segments
- Teachers = routers / switches
- Messages between students = data packets

When two students want to talk, they need a path (like a line of communication). In computers, that path is **cables, Wi-Fi, routers, switches**, etc.

Key goals of networking:

- Share files, printers, internet
- Send messages, emails, videos
- Access websites, cloud services

- Talk to other networks (like the internet)

All the topics you listed (OSI, TCP/IP, IP, subnetting, routing, switching, NAT, VLANs, DNS, DHCP) are pieces of this big puzzle.

## 2. OSI Model – The 7-Layer “Language” of Networks

The **OSI model** (Open Systems Interconnection) is a **theoretical 7-layer map** that describes how computers talk to each other over a network.

It is **not a physical thing**, but a **conceptual model** that helps us:

- Understand how data moves
- Decide where problems are (which layer)
- Design networks in an organized way
- Standardize devices and protocols

Think of OSI like a **package delivery system** from your house to a friend’s house:

- You write a letter (Application)
- Put it in an envelope with address (Network)
- Put envelope in a box (Data Link)
- Truck takes it on roads (Physical)
- Delivery person gives it to your friend (all layers together)

### 2.1 The 7 Layers (Top to Bottom)

Mnemonic: **“All People Seem To Need Data Processing”**  
(7 → 1)

| Layer | Name         | Main Job                              | Real-life Analogy                                |
|-------|--------------|---------------------------------------|--------------------------------------------------|
| 7     | Application  | What you see and use (apps, browser)  | You opening WhatsApp and typing a message        |
| 6     | Presentation | Format, encrypt, compress data        | Writing in a language friend understands         |
| 5     | Session      | Start, manage, end sessions           | Keeping WhatsApp chat open without reconnecting  |
| 4     | Transport    | Reliable delivery (TCP) or fast (UDP) | Ensuring message arrives fully or fast live call |
| 3     | Network      | Logical addressing & routing (IP)     | Using GPS to find friend’s house address         |

| Layer | Name      | Main Job                        | Real-life Analogy                                    |
|-------|-----------|---------------------------------|------------------------------------------------------|
| 2     | Data Link | Direct connection (MAC, frames) | Ensuring message goes to right device on same street |
| 1     | Physical  | Wires, signals, bits            | Cables, Wi-Fi waves, electricity in wires            |

## 2.2 Layer-by-Layer Deep Explanation with Example

**Scenario:** You send a **WhatsApp message** to your friend.

### Layer 7 – Application Layer

- This is the **layer you interact with directly**.
- Examples: Web browser (Chrome), WhatsApp, Gmail, YouTube app.
- Protocols: HTTP, HTTPS, FTP, SMTP, DNS, etc.

#### What happens:

You open WhatsApp and type: “Hello, how are you?”

This text is the **application data**.

#### Simple child view:

This is you **typing the message** on your phone.

### Layer 6 – Presentation Layer

- Translates data into a **common format** both devices understand.
- Handles **encryption** (security), **compression**, and **format conversion**.

#### What happens:

WhatsApp **encrypts** your message so others cannot read it. It may also compress images/videos.

#### Simple child view:

You put your message in a **secret code** that only your friend can understand.

### Layer 5 – Session Layer

- Manages **sessions** (connections) between applications.
- Starts, keeps alive, and ends the session.

#### What happens:

WhatsApp stays **logged in** and keeps the chat session open so you don’t have to reconnect every time you send a message.

### Simple child view:

You and your friend stay on the **same phone call** instead of redialing for each sentence.

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### Layer 4 – Transport Layer

- Breaks data into **segments**.
- Ensures **reliable delivery** (TCP) or **fast delivery** (UDP).
- Uses **port numbers** (like door numbers) to direct data to the right app.

### Key protocols:

- **TCP** (Transmission Control Protocol): Reliable, checks for errors, resends lost data. Used for web pages, emails, files.
- **UDP** (User Datagram Protocol): Fast, no guarantee. Used for live video, gaming, voice calls.

### What happens:

Your message is split into small pieces (segments). TCP ensures all pieces arrive correctly and in order. Port 443 (HTTPS) or WhatsApp's port is added.

### Simple child view:

Your long letter is cut into **small envelopes**, each labeled with an order number so your friend can reassemble them correctly.

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### Layer 3 – Network Layer

- Adds **logical addresses** (IP addresses).
- Decides **how to route** packets from source to destination across many networks.
- Works with **routers**.

### What happens:

Your device adds:

- Source IP: your IP (e.g., 192.168.1.10)
- Destination IP: WhatsApp server's IP (e.g., 142.250.x.x)

Routers use these IPs to find the best path through the internet.

### Simple child view:

You write **your address** and **friend's full address** on the envelope so the postal service knows where to send it.

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### Layer 2 – Data Link Layer

- Adds **MAC addresses** (physical hardware addresses).
- Creates **frames**.

- Works within the **same local network** (same LAN).
- Used by **switches**.

**What happens:**

Your device adds:

- Source MAC: your device's MAC address
- Destination MAC: your router's MAC address (next hop)

The frame is sent over your local network to the router.

**Simple child view:**

You write the **street name and house number** for the local post office that will take your letter to the bigger city.

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**Layer 1 – Physical Layer**

- Deals with **actual bits** (0s and 1s) as **electrical signals, light, or radio waves**.
- Cables (Ethernet, fiber), Wi-Fi signals, connectors, voltages.

**What happens:**

The frame is converted into **electrical signals** on Ethernet cable or **radio waves** over Wi-Fi.

**Simple child view:**

The letter is put on a **truck** and travels on **roads** to the next city.

## 2.3 Encapsulation and Decapsulation

When you send data:

- Each layer **adds its own header** (like putting a letter in envelopes inside envelopes). This is **encapsulation**.
- When data arrives, each layer **removes its header** in reverse order. This is **decapsulation**.

Example (sending):

1. Application data
2. Presentation header
3. Session header
4. Transport header (TCP) → segment
5. Network header (IP) → packet
6. Data Link header & trailer (MAC) → frame
7. Bits on wire (Physical)



At the receiver side, this is reversed.

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### 3. TCP/IP Model – The Real-World Internet Model

The **TCP/IP model** is the **practical model** actually used on the internet. It has **4 layers** instead of 7 and is based on real protocols (TCP, IP, etc.).

Think of OSI as a **detailed textbook**, and TCP/IP as the **actual working system** on the internet.

#### 3.1 TCP/IP Layers and Mapping to OSI

| TCP/IP Layer   | OSI Layers it Covers                           | Main Job                          |
|----------------|------------------------------------------------|-----------------------------------|
| Application    | 5 (Session), 6 (Presentation), 7 (Application) | Apps, protocols (HTTP, DNS, DHCP) |
| Transport      | 4 (Transport)                                  | TCP, UDP, ports, reliability      |
| Internet       | 3 (Network)                                    | IP addressing, routing            |
| Network Access | 2 (Data Link) + 1 (Physical)                   | Ethernet, Wi-Fi, MAC, cables      |

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#### 3.2 Simple Real-Life Analogy: Ordering Food on Swiggy/Zomato

**Scenario:** You order food using an app.

- **Network Access Layer:**  
Your phone connects to the internet via **Wi-Fi or mobile data** (cables, signals).
  - **Internet Layer:**  
Your phone finds the **restaurant's IP address** (like finding restaurant location).
  - **Transport Layer:**  
The delivery person picks up food and ensures it arrives **safely and in order** (like TCP).
  - **Application Layer:**  
You open the app, see the food, and **enjoy eating**. This is what you directly interact with.
- 

#### 3.3 Key Protocols in TCP/IP

- **TCP:** Reliable, connection-oriented, ensures full, correct delivery.  
Example: Loading a webpage, downloading a file.
- **UDP:** Fast, no guarantee, no retransmission.  
Example: Live video call, online gaming, streaming.
- **IP:** Adds source and destination IP addresses, routes packets.

- Application protocols: HTTP, HTTPS, FTP, DNS, DHCP, SMTP, etc.

#### 4. IPv4 – The Classic Internet Address (32-bit)

An **IP address** is like a **house address** for a device on a network. It tells other devices **where to send data**.

**IPv4** uses **32-bit** addresses, written in **dotted decimal** format.

Example: 192.168.1.10

##### 4.1 IPv4 Address Structure

- 32 bits = 4 groups of 8 bits (octets).
- Each octet is 0–255 in decimal.

Example:

192.168.1.10 in binary:

11000000.10101000.00000001.00001010

##### Network Part vs Host Part

- Part of the address identifies the **network** (like street name).
- Remaining part identifies the **host** (like house number).

Example:

In 192.168.1.10/24:

- Network: 192.168.1.0
- Host: .10

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##### 4.2 IPv4 Address Classes (Old but Good to Know)

Classful addressing divides IPv4 into classes:

| Class | First Octet Range | Network/Host Bits   | Example Use           |
|-------|-------------------|---------------------|-----------------------|
| A     | 1 – 126           | 8 network, 24 host  | Very large networks   |
| B     | 128 – 191         | 16 network, 16 host | Medium networks       |
| C     | 192 – 223         | 24 network, 8 host  | Small networks (home) |
| D     | 224 – 239         | Multicast           | One-to-many streaming |

| Class | First Octet Range | Network/Host Bits | Example Use           |
|-------|-------------------|-------------------|-----------------------|
| E     | 240 – 255         | Reserved          | Research / future use |

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### 4.3 Public vs Private IPv4 Addresses

#### Public IP

- Unique on the **entire internet**.
- Assigned by your ISP.
- Example: 203.0.113.10

#### Private IP

- Used **inside local networks** (home, office).
- Not routable on the public internet.
- Many networks can use the same private IPs.

#### Private ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Your home router usually gives devices addresses like 192.168.1.2, 192.168.1.3, etc.

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### 4.4 Special IPv4 Addresses

- 0.0.0.0 – Default/unspecified address
- 255.255.255.255 – Limited broadcast (all devices on local network)
- 127.0.0.1 – Loopback (your own computer, “localhost”)
- 169.254.x.x – APIPA (auto-assigned when DHCP fails)

## 5. IPv6 – The New Internet Address (128-bit)

IPv4 addresses are almost exhausted (only ~4.3 billion).

**IPv6** uses **128-bit** addresses, giving an **enormous** number of addresses.

### 5.1 IPv6 Address Format

- Written in **hexadecimal**, 8 groups of 4 hex digits, separated by colons.

Example full form:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Shortened rules:

- Leading zeros in each group can be removed:  
0db8 → db8, 0370 → 370
- One连续 block of all zeros can be replaced with :: (only once).

Short form:

2001:db8:85a3::8a2e:370:7334

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## 5.2 IPv6 vs IPv4 – Key Differences

| Feature          | IPv4                         | IPv6                          |
|------------------|------------------------------|-------------------------------|
| Address size     | 32 bits                      | 128 bits                      |
| Notation         | Decimal dotted (192.168.1.1) | Hex colon (2001:db8::1)       |
| Address count    | ~4.3 billion                 | ~ $3.4 \times 10^{38}$ (huge) |
| Configuration    | Manual or DHCP               | SLAAC, DHCPv6, manual         |
| NAT              | Often needed                 | Usually not needed            |
| Security (IPsec) | Optional                     | Built-in support              |
| Broadcast        | Yes                          | No (uses multicast/anycast)   |
| Header checksum  | Yes                          | No                            |
| Fragmentation    | Sender + routers             | Sender only                   |

### Why IPv6 is Better

- **No address shortage:** Every device can have a unique global IP.
- **Simpler routing:** Larger, cleaner address space.

- **Better security:** IPsec built in.
- **No NAT needed:** End-to-end connectivity restored.

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## 6. Subnetting – Dividing a Big Network into Smaller Ones

**Subnetting** is the process of dividing a large IP network into **smaller logical networks** called **subnets**.

Think of a big **city** divided into **neighborhoods**. Each neighborhood is easier to manage.

### 6.1 Why Subnet?

- **Reduce broadcast traffic:** Broadcasts stay within a subnet.
- **Improve security:** Separate departments (HR, IT, Sales).
- **Better organization:** Easier to manage and troubleshoot.
- **Efficient IP usage:** Don't waste addresses.

---

### 6.2 Subnet Mask and CIDR Notation

#### Subnet Mask

A **subnet mask** tells which part of an IP is **network** and which is **host**.

Example:

- IP: 192.168.1.10
- Subnet mask: 255.255.255.0

This means:

- First 3 octets (192.168.1) = network part
- Last octet (.10) = host part

#### CIDR Notation

Instead of writing 255.255.255.0, we write /24.

- /24 means **24 bits** are for network, remaining 8 bits for hosts.
- Example: 192.168.1.0/24

Common CIDR values:

| CIDR | Subnet Mask | Hosts per subnet (usable) |
|------|-------------|---------------------------|
| /8   | 255.0.0.0   | ~16 million               |

| CIDR | Subnet Mask     | Hosts per subnet (usable) |
|------|-----------------|---------------------------|
| /16  | 255.255.0.0     | ~65,000                   |
| /24  | 255.255.255.0   | 254                       |
| /25  | 255.255.255.128 | 126                       |
| /26  | 255.255.255.192 | 62                        |

### 6.3 Step-by-Step Subnetting Example

**Goal:** You have 192.168.1.0/24 and need **2 subnets** (Office and Lab).

Original:

- Network: 192.168.1.0/24
- Hosts: 254 usable (.1 to .254)

To get 2 subnets, borrow **1 bit** from host part:

- New prefix: /25
- Subnets: 2
  - 192.168.1.0/25 → hosts .1–.126
  - 192.168.1.128/25 → hosts .129–.254

You can assign:

- Office: 192.168.1.0/25
- Lab: 192.168.1.128/25

Now they are **logically separated**. Broadcasts in one subnet do not affect the other.

### 6.4 Network Address, Broadcast Address, Usable Hosts

For any subnet:

- **Network address:** First address (all host bits 0). Cannot be assigned to a device.
- **Broadcast address:** Last address (all host bits 1). Used to send to all devices in subnet.
- **Usable hosts:** All addresses between network and broadcast.

Example for 192.168.1.0/24:

- Network: 192.168.1.0
- Broadcast: 192.168.1.255
- Usable hosts: 192.168.1.1 – 192.168.1.254 (254 addresses)

## 7. Routing – Sending Packets Between Networks

**Routing** is the process of choosing a **path** for packets to travel from source network to destination network.

**Routers** are devices that perform routing.

Think of routing like **GPS navigation** for data:

- Your destination = IP address of server
- Roads = networks and links
- Router = GPS that decides which road to take

### 7.1 What Routers Do

- Connect **multiple networks** (e.g., your home LAN and ISP network).
- Maintain a **routing table** (map of networks and paths).
- Look at **destination IP** and decide **next hop**.
- Forward packets between networks.
- Drop packets if no route exists.

### 7.2 Routing Table (Simple View)

A routing table might look like:

| Destination Network | Subnet Mask   | Next Hop / Interface          |
|---------------------|---------------|-------------------------------|
| 192.168.1.0         | 255.255.255.0 | directly connected (eth0)     |
| 0.0.0.0             | 0.0.0.0       | 203.0.113.1 (default gateway) |

- 0.0.0.0/0 = **default route**: “if no better match, send here.”

### 7.3 Static Routing vs Dynamic Routing

#### Static Routing

- Routes are **manually configured** by an admin.
- Fixed path; does not change automatically.
- Simple, secure, low CPU usage.
- Good for small networks.

**Analogy:** You know a **fixed path** to a store and always walk the same way, even if there is traffic.

### Dynamic Routing

- Routes are learned **automatically** using **routing protocols**.
- Protocols: RIP, OSPF, EIGRP, BGP.
- Adapts to network changes (link failures, congestion).
- Used in large networks and the internet.

**Analogy:** Google Maps **recomputes route** based on traffic, accidents, etc.

---

## 7.4 Default Gateway

In your home network:

- Your PC: 192.168.1.10
- Router (gateway): 192.168.1.1

When your PC wants to reach an IP outside its subnet (e.g., 93.184.216.34), it sends the packet to the **default gateway** (router). The router then forwards it toward the internet.

---

## 8. Switching – Forwarding Frames Inside a LAN

**Switching** is the process of forwarding **frames** within a **local network** based on **MAC addresses**.

Switches work mainly at **OSI Layer 2 (Data Link)**.

Think of a switch as a **smart postal office** inside a building that knows exactly which room each person lives in.

---

### 8.1 MAC Address – The Physical Address

- **MAC** = Media Access Control address.
- Unique hardware address burned into NIC (network card).
- Format: 6 bytes (12 hex digits), e.g., 00:1A:2B:3C:4D:5E.

Each device has a **unique MAC** worldwide (in theory).

---

### 8.2 How a Switch Learns MAC Addresses



A switch builds a **MAC address table**:

- It looks at the **source MAC** of each incoming frame.
- It records: "This MAC is on this port."

Example:

- PC1 (MAC AAA) on port 1 sends a frame.
  - Switch learns: AAA → port 1.
  - PC2 (MAC BBB) on port 5 sends a frame.
  - Switch learns: BBB → port 5.
- 

### 8.3 Forwarding Decision

When a frame arrives:

1. Switch checks **destination MAC**.
2. If MAC is in the table:
  - Forward frame **only** to that port.
3. If MAC is **not** in the table:
  - **Flood** frame out all ports except the incoming one.
4. When destination replies, switch learns its MAC.

Example:

- PC1 sends to PC2:
    - Switch doesn't know PC2 yet → floods to all ports.
    - PC2 receives and replies.
    - Switch learns PC2's MAC on port 5.
    - Next time, switch sends directly to port 5.
- 

### 8.4 Hub vs Switch

| Feature      | Hub                       | Switch                            |
|--------------|---------------------------|-----------------------------------|
| Intelligence | None (dumb)               | Smart (learns MAC table)          |
| Forwarding   | Sends to <b>all</b> ports | Sends to <b>only</b> correct port |

| Feature     | Hub             | Switch                       |
|-------------|-----------------|------------------------------|
| Collisions  | Many collisions | Few/no collisions (per port) |
| Performance | Low             | High                         |

Switches are much better than hubs.

---

## 8.5 Collision Domain vs Broadcast Domain

### Collision Domain

- A **collision domain** is a network segment where **data collisions** can happen.
- In a **hub**, all devices share one collision domain.
- In a **switch**, **each port** is a separate collision domain.

### Broadcast Domain

- A **broadcast domain** is the set of devices that receive **broadcast frames** (destination MAC = FF:FF:FF:FF:FF:FF).
  - By default, all ports on a switch are in the **same broadcast domain**.
  - **VLANs** split a switch into multiple broadcast domains.
- 

## 9. NAT – Network Address Translation

**NAT** allows many devices with **private IP addresses** to share **one public IP** when accessing the internet.

Think of an **office building** with one front door (public IP). All employees (private IPs) use that door to go outside. When replies come back, the receptionist (NAT) knows which employee they belong to.

---

### 9.1 Why NAT Is Used

- **IPv4 address shortage**: Many devices share one public IP.
  - **Security**: Internal addresses are hidden from the internet.
  - **Simplifies network design**: Internal network can use private IPs freely.
- 

### 9.2 How NAT Works (Home Router Example)

Your home network:

- Public IP (from ISP): 203.0.113.10
- Private IPs:
  - Phone: 192.168.1.2
  - Laptop: 192.168.1.3
  - PC: 192.168.1.4

When your **laptop** (192.168.1.3) requests a website:

1. Laptop sends packet:
  - Source: 192.168.1.3
  - Destination: web server IP (e.g., 93.184.216.34)
2. Router (NAT) sees this and:
  - Replaces source IP with its public IP 203.0.113.10.
  - Adds a **port number** to track this connection.
  - Stores mapping in **NAT table**.
3. Web server replies to 203.0.113.10.
4. Router checks NAT table, sees this reply belongs to 192.168.1.3, and forwards it.

---

### 9.3 NAT Table (Simplified)

Example NAT table entry:

| Local IP    | Local Port | Public IP    | Public Port | Remote IP     | Remote Port |
|-------------|------------|--------------|-------------|---------------|-------------|
| 192.168.1.3 | 5000       | 203.0.113.10 | 60000       | 93.184.216.34 | 80          |

router uses this to map replies back correctly.

---

### 9.4 Types of NAT (Brief)

- **Static NAT:** One private IP ↔ one public IP (fixed mapping).
  - **Dynamic NAT:** Pool of public IPs, assigned dynamically.
  - **PAT (Port Address Translation) / NAT Overload:** Many private IPs share one public IP using different ports (most common in homes).
-

## 10. VLANs – Virtual LANs (Logical Network Segmentation)

A **VLAN (Virtual Local Area Network)** is a **logical group** of devices on a switch (or multiple switches) that behave like they are on the same physical LAN, even if they are not physically together.

Think of a **big school building** where:

- All students are in one building (one switch).
- But they are divided into **sections** (VLANs): Section A, Section B, Section C.
  - Section A = HR department
  - Section B = IT department
  - Section C = Sales department

Each section can talk within itself easily, but to talk between sections, they need a **passage** (router).

---

### 10.1 Why Use VLANs?

- **Segmentation:** Separate departments, users, or services.
  - **Security:** Limit access between groups.
  - **Reduce broadcast traffic:** Each VLAN is its own broadcast domain.
  - **Flexibility:** Logical grouping independent of physical location.
- 

### 10.2 VLAN Basics

- Each VLAN has a **VLAN ID** (e.g., VLAN 10, VLAN 20).
  - Switch ports are assigned to a VLAN.
  - Devices on the same VLAN can communicate directly (Layer 2).
  - Devices on different VLANs **cannot** communicate directly; they need a **router** or **Layer 3 switch** (inter-VLAN routing).
- 

### 10.3 Example: Office VLANs

On a single switch:

- VLAN 10 – HR
  - Ports: 1–8
- VLAN 20 – IT
  - Ports: 9–16
- VLAN 30 – Sales
  - Ports: 17–24

All HR PCs are in VLAN 10, even if they are on different floors.  
They behave like they are on the same physical LAN.

Broadcasts from HR stay in VLAN 10 and do not reach IT or Sales.

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#### 10.4 Inter-VLAN Routing

To allow HR (VLAN 10) to talk to IT (VLAN 20):

- Use a **router** or **Layer 3 switch**.
- Two common methods:
  1. **Router on a stick**: One router interface with multiple sub-interfaces (one per VLAN).
  2. **SVI (Switch Virtual Interface)** on a Layer 3 switch.

Think of it as a **bridge** between different sections of the school; only authorized people can cross.

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### 11. DNS Fundamentals – The Internet Phonebook

**DNS (Domain Name System)** translates **human-readable domain names** (like google.com) into **IP addresses** (like 142.250.x.x) that computers use.

Without DNS, you would have to remember IPs like 142.250.185.78 for Google.

Think of DNS as a **phonebook**:

- Name = domain name (google.com)
  - Phone number = IP address (142.250.x.x)
- 

#### 11.1 Why DNS Is Needed

- Humans remember names better than numbers.
  - Servers can change IP addresses; DNS can be updated without users noticing.
  - One domain can map to multiple IPs (for load balancing).
- 

#### 11.2 How DNS Works – Step by Step

**Scenario:** You type www.example.com in your browser.

1. **Client checks local cache**
  - Your computer checks if it already knows the IP for www.example.com.
  - If yes, use it immediately.
  - If no, ask your **DNS resolver** (usually your router or ISP DNS like 8.8.8.8 or 1.1.1.1).
2. **Resolver contacts Root DNS Server**

- Root server says: “For .com, ask this TLD server.”
- There are 13 sets of root DNS servers globally.

### 3. Resolver contacts TLD (.com) Server

- TLD server says: “For example.com, ask this authoritative nameserver.”

### 4. Resolver contacts Authoritative DNS Server

- This server holds the real DNS records for example.com.
- It returns the IP address, e.g., 93.184.216.34.

### 5. Resolver returns IP to your computer

- Your computer caches the result for some time.
- Browser connects to 93.184.216.34 and loads the website.

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## 11.3 Common DNS Record Types

| Record Type | Purpose                               | Example                        |
|-------------|---------------------------------------|--------------------------------|
| A           | Map domain to IPv4 address            | example.com → 93.184.216.34    |
| AAAA        | Map domain to IPv6 address            | example.com → 2606:2800::1     |
| CNAME       | Alias (domain → another domain)       | www → example.com              |
| MX          | Mail server for email                 | example.com → mail.example.com |
| TXT         | Text record (SPF, verification, etc.) | SPF for email security         |

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## 12. DHCP Fundamentals – Automatic IP Configuration

**DHCP (Dynamic Host Configuration Protocol)** automatically gives devices **IP addresses and network settings** when they join a network.

Without DHCP, you would have to manually configure every device with:

- IP address
- Subnet mask
- Default gateway
- DNS server

Think of DHCP as a **receptionist** that gives each new visitor an ID card with all necessary details.

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## 12.1 What DHCP Provides

When a device connects:

- IP address (e.g., 192.168.1.10)
  - Subnet mask (e.g., 255.255.255.0)
  - Default gateway (e.g., 192.168.1.1)
  - DNS server(s) (e.g., 8.8.8.8, 8.8.4.4)
  - Lease time (how long the IP is valid)
- 

## 12.2 DHCP DORA Process – Step by Step

**DORA = Discover, Offer, Request, Acknowledge**

### Step 1 – Discover

- New device (client) connects to network.
- It doesn't have an IP yet.
- It broadcasts: **"Is there any DHCP server?"**  
(DHCP Discover message)

### Step 2 – Offer

- DHCP server receives the Discover.
- It picks an available IP from its pool.
- It sends a **DHCP Offer**:
  - Proposed IP
  - Subnet mask
  - Gateway
  - DNS
  - Lease time

This is also broadcast (since client doesn't have an IP yet).

### Step 3 – Request

- Client chooses one offer (usually the first).
- It broadcasts: **"I want to use this IP."**  
(DHCP Request)

### Step 4 – Acknowledge

- DHCP server confirms: **“OK, you can use this IP.”**  
(DHCP ACK)
- Client now configures its interface with the IP and settings.

If the IP is no longer available, the server sends a **NACK** (negative acknowledgment), and the client starts again.

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### 12.3 DHCP Lease and Renewal

- IP is assigned for a **lease time** (e.g., 24 hours).
  - Before lease expires:
    - Client tries to **renew** with the same server.
    - If successful, lease is extended.
  - If lease expires and not renewed, IP returns to the pool.
- 

## 13. Putting Everything Together – End-to-End Example

**Scenario:** Siddhesh’s laptop connects to Wi-Fi and opens example.com.

### Step 1 – Physical & Data Link (Wi-Fi, MAC)

- Laptop connects to Wi-Fi (Physical + Data Link).
  - Laptop has a MAC address, e.g., 00:1A:2B:3C:4D:5E.
  - Router’s Wi-Fi MAC is known.
- 

### Step 2 – DHCP (Get IP)

- Laptop has no IP yet.
  - DHCP DORA process runs:
    - Discover → Offer → Request → ACK
  - Laptop gets:
    - IP: 192.168.1.10
    - Mask: 255.255.255.0
    - Gateway: 192.168.1.1
    - DNS: 8.8.8.8
- 

### Step 3 – DNS (Get IP for example.com)

- Laptop wants to open example.com.



- It asks DNS resolver (8.8.8.8) for example.com.
  - DNS resolves to IP: 93.184.216.34.
- 

#### **Step 4 – TCP Connection (Transport Layer)**

- Laptop creates a TCP connection to 93.184.216.34:80 (HTTP) or :443 (HTTPS).
  - TCP ensures reliable delivery.
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#### **Step 5 – Network Layer (IP) and Routing**

- Laptop creates an IP packet:
    - Source IP: 192.168.1.10
    - Destination IP: 93.184.216.34
  - Since destination is not in local network, laptop sends packet to **default gateway** (192.168.1.1, router).
- 

#### **Step 6 – NAT on Router**

- Router sees packet from 192.168.1.10 to internet.
  - Router performs **NAT**:
    - Replaces source IP with its public IP 203.0.113.10.
    - Adds port mapping in NAT table.
  - Sends packet to ISP, then to internet.
- 

#### **Step 7 – Routing Across the Internet**

- Many routers along the internet use their routing tables to forward packets toward 93.184.216.34.
  - Each router decides next hop based on destination IP.
- 

#### **Step 8 – Server Responds**

- Server at 93.184.216.34 sends response back to 203.0.113.10.
  - Router receives reply, checks NAT table, sees it belongs to 192.168.1.10.
  - Router translates destination IP back to 192.168.1.10 and forwards to laptop.
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#### **Step 9 – Switch and LAN**

- Router sends frame to switch.
- Switch looks at MAC address table and forwards frame to laptop's port.
- Laptop receives TCP segments, reassembles data, browser shows example.com.

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#### Step 10 – OSI Layers in This Flow

- **Application:** Browser, HTTP/HTTPS.
  - **Presentation:** Encryption (HTTPS), data format.
  - **Session:** TCP session between laptop and server.
  - **Transport:** TCP segments, ports.
  - **Network:** IP packets, routing.
  - **Data Link:** Ethernet/Wi-Fi frames, MAC addresses.
  - **Physical:** Wi-Fi signals, cables.
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#### 14. Quick Summary Table of All Topics

| Topic        | What It Is                                         | Key Idea in One Line                    |
|--------------|----------------------------------------------------|-----------------------------------------|
| Topic        | What It Is                                         | Key Idea in One Line                    |
| OSI Model    | 7-layer conceptual model of networking             | How data moves layer by layer           |
| TCP/IP Model | 4-layer practical internet model                   | Real model used on the internet         |
| IPv4         | 32-bit IP address (e.g., 192.168.1.10)             | Classic internet address                |
| IPv6         | 128-bit IP address (e.g., 2001:db8::1)             | New, huge address space                 |
| Subnetting   | Dividing a network into smaller subnets            | Smaller, safer, more efficient networks |
| Routing      | Moving packets between networks using routers      | GPS for data across networks            |
| Switching    | Forwarding frames inside a LAN using MAC addresses | Smart local delivery within a building  |

| Topic | What It Is                                     | Key Idea in One Line                         |
|-------|------------------------------------------------|----------------------------------------------|
| NAT   | Translating private IPs to public IP           | Many devices share one public IP             |
| VLANs | Logical network segments on a switch           | Separate departments on same physical switch |
| DNS   | Translates domain names to IP addresses        | Internet phonebook                           |
| DHCP  | Automatically gives IP and settings to devices | Auto ID card for new devices on network      |